

Polymerisation

A Level Chemistry

Addition polymerisation

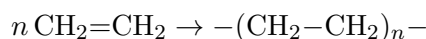
In **addition polymerisation** 加成聚合, many small molecules join into one very long chain, with no other product made.

Each small molecule is a **monomer** 单体. It must be unsaturated—it has a C=C double bond. The double bond opens up so that the monomers can link together. The long chain that forms is the **polymer** 聚合物.

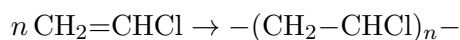
Repeat units

The **repeat unit** 重复单元 is the small part that is copied again and again along the chain. To find it, take the monomer, change the C=C to a single C–C, and draw bonds going out at each end.

- **poly(ethene)** 聚乙烯 is made from ethene:



- **poly(chloroethene)** 聚氯乙烯 (PVC) is made from chloroethene:



Finding the monomer

To go the other way, look at one repeat unit of the polymer, and put the C=C double bond back in. That gives you the monomer.

The problem of disposal

Poly(alkene)s are very hard to get rid of:

- they are **non-biodegradable** 不可生物降解—microbes cannot break them down, so they stay in the ground for a very long time.
- their **combustion** 燃烧 (burning) can release harmful gases. For example, burning PVC gives off toxic hydrogen chloride.